

PORT ARTHUR, TX, USA

WMO: 722410

Lat: 29.951N

Lon: 94.021W

Elev: 5

StdP: 101.27

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.3	(c) 1.6	(d) -8.2	(e) 1.9	(f) 4.0	(g) -5.4	(h) 2.4	(i) 6.5	(j) 11.0	(k) 14.0	(l) 10.0	(m) 13.4	(n) 3.5	(o) 20	(p) 0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.2	(c) 35.0	(d) 25.5	(e) 34.1	(f) 25.6	(g) 33.3	(h) 25.5	(i) 27.5	(j) 31.8	(k) 26.9	(l) 31.2	(m) 26.7	(n) 30.9	(o) 3.5	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	29.3	25.9	21.3	28.9	25.6	20.8	28.5	86.8	31.5	84.7	31.4	82.9	30.9	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.7	7.9	DB	-3.1	37.0	1.8	1.5	-4.4	38.1	-5.4	39.0	-6.4	39.9	-7.7	41.0
			WB	-4.7	28.4	1.7	1.1	-6.0	29.2	-7.0	29.8	-8.0	30.4	-9.2	31.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	11.9	14.1	17.2	20.6	24.7	27.7	28.5	28.6	26.5	21.9	16.7	13.0
	DBStd	7.06	5.30	5.14	4.41	3.45	2.49	1.59	1.26	1.52	2.40	3.93	4.67	5.21
	HDD10.0	98	41	19	5	0	0	0	0	0	0	0	5	28
	HDD18.3	714	207	135	74	18	1	0	0	0	0	15	84	181
	CDD10.0	4104	101	133	228	319	456	529	572	576	496	368	207	120
	CDD18.3	1680	8	15	39	87	198	279	314	318	246	124	36	14
	CDH23.3	15726	14	33	134	484	1650	2957	3467	3488	2323	962	176	39
	CDH26.7	6131	0	1	8	66	496	1257	1524	1574	923	264	18	1
Wind	WSAvg	3.7	4.3	4.5	4.5	4.5	4.1	3.4	2.7	2.6	3.1	3.5	3.8	4.1
Precipitation	PrecAvg	1476	131	90	84	93	132	152	129	148	159	117	121	122
	PrecMax	2072	378	195	238	389	335	480	398	438	558	383	275	457
	PrecMin	875	15	5	5	7	0	19	38	22	25	0	4	49
	PrecStd	304	79	52	54	74	95	108	71	108	115	98	69	79
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	26.1	27.8	29.9	33.3	35.5	35.7	36.9	35.6	32.3	28.5	26.2
		MCWB	19.2	19.8	20.5	21.8	23.5	24.7	25.5	25.3	24.8	24.3	22.9	21.5
	2%	DB	23.3	24.3	26.2	28.3	31.4	34.0	34.5	35.2	33.8	30.9	26.9	24.4
		MCWB	18.9	19.8	20.2	21.6	23.7	25.1	25.8	25.8	24.8	23.9	21.8	21.0
	5%	DB	21.8	22.8	24.9	27.3	30.3	33.0	33.6	34.1	32.5	29.5	25.4	22.8
		MCWB	18.4	19.0	19.8	21.2	23.7	25.2	25.9	25.9	24.7	23.0	20.7	20.1
	10%	DB	20.1	21.4	23.5	26.1	29.4	32.1	32.8	33.1	31.4	28.1	23.9	21.2
		MCWB	17.5	18.5	19.4	20.9	23.5	25.2	25.9	25.9	24.4	22.4	20.2	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.9	22.7	24.8	26.9	28.0	28.0	28.1	27.3	26.5	24.3	23.0
		MCDB	23.1	24.4	25.3	27.3	29.6	31.6	32.7	32.4	30.5	29.3	26.7	24.7
	2%	WB	20.5	21.2	21.9	23.8	25.8	26.9	27.3	27.5	26.6	25.7	23.2	21.8
		MCDB	22.1	23.1	24.2	26.2	28.7	30.8	32.0	31.9	30.4	28.5	25.4	23.6
	5%	WB	19.6	20.3	21.3	23.1	25.1	26.5	26.8	26.9	26.1	24.9	22.3	20.9
		MCDB	21.1	21.9	23.3	25.5	28.4	30.4	31.4	31.3	30.0	27.6	24.4	22.4
	10%	WB	18.2	19.4	20.6	22.5	24.5	26.0	26.5	26.6	25.6	24.1	21.3	19.5
		MCDB	19.6	20.8	22.5	24.7	27.8	29.9	30.9	31.0	29.3	26.7	23.3	20.8
Mean Daily Temperature Range	5% DB	MDBR	10.1	9.7	9.9	9.8	9.1	8.8	8.9	9.2	9.4	10.5	10.6	9.9
		MCDBR	10.7	9.8	9.6	9.9	9.6	10.1	10.1	10.6	10.4	10.4	10.5	10.2
		MCWBR	7.2	6.1	5.1	4.8	3.9	3.3	3.4	3.3	3.6	4.6	6.0	7.1
	5% WB	MCDBR	8.6	8.3	8.3	7.5	7.8	8.4	9.1	9.3	8.4	8.3	9.3	9.0
		MCWBR	6.8	6.0	5.4	4.6	3.8	3.5	3.6	3.5	3.4	4.7	6.6	7.1
Clear-Sky Solar Irradiance	taub	0.342	0.358	0.376	0.433	0.478	0.491	0.501	0.482	0.457	0.385	0.372	0.351	
	taud	2.518	2.482	2.424	2.263	2.170	2.179	2.186	2.242	2.278	2.464	2.441	2.496	
	Ebn at Noon	893	907	910	865	824	808	799	811	820	866	848	862	
	Edh at Noon	89	100	113	137	151	149	148	138	129	101	95	87	
All-Sky Solar Radiation	RadAvg	2.88	3.25	4.52	5.44	6.11	6.26	5.99	5.67	4.99	4.38	3.31	2.54	
	RadStd	0.28	0.54	0.40	0.39	0.41	0.46	0.50	0.48	0.51	0.55	0.33	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	+0.37	N/A	-0.35	N/A	N/A	N/A	+96
Regional (33 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+90

Nomenclature: See separate page